

Q1.

$$z = 3 - i$$

Determine the value of zz^*

Circle your answer.

10 $\sqrt{10}$ $10 - 2i$ $10 + 2i$

(Total 1 mark)

Q2.

- (a) Solve the equation $w^2 + 6w + 34 = 0$, giving your answers in the form $p + qi$, where p and q are integers.

(3)

- (b) It is given that $z = i(1 + i)(2 + i)$.

- (i) Express z in the form $a + bi$, where a and b are integers.

(3)

- (ii) Find integers m and n such that $z + mz^* = ni$.

(3)

(Total 9 marks)

Q3.

- (a) It is given that $z = x + yi$, where x and y are real numbers.

- (i) Write down, in terms of x and y , an expression for $(z - 2i)^*$.

(1)

- (ii) Solve the equation

$$(z - 2i)^* = 4iz + 3$$

giving your answer in the form $a + bi$.

(5)

- (b) It is given that $p + qi$, where p and q are real numbers, is a root of the equation $z^2 + 10iz - 29 = 0$.

Without finding the values of p and q , **state** why $p - qi$ is **not** a root of the equation $z^2 + 10iz - 29 = 0$.

(1)

(Total 7 marks)

Q4.

The cubic equation

$$z^3 + pz^2 + qz + 37 - 36i = 0$$

where p and q are constants, has three complex roots, α , β and γ . It is given that $\beta = -2 + 3i$ and $\gamma = 1 + 2i$.

- (a) (i) Write down the value of $\alpha\beta\gamma$. (1)
- (ii) Hence show that $(8 + i)\alpha = 37 - 36i$. (2)
- (iii) Hence find α , giving your answer in the form $m + ni$, where m and n are integers. (3)
- (b) Find the value of p . (1)
- (c) Find the value of the complex number q . (2)
- (Total 9 marks)**

Q5.

- (a) Solve the following equations, giving each root in the form $a + bi$:
- (i) $x^2 + 9 = 0$; (1)
- (ii) $(x + 2)^2 + 9 = 0$. (1)
- (b) (i) Expand $(1 + x)^3$. (1)
- (ii) Express $(1 + 2i)^3$ in the form $a + bi$. (3)
- (iii) Given that $z = 1 + 2i$, find the value of $z^* - z^3$. (2)
- (Total 8 marks)**

Q6.

The numbers α , β and γ satisfy the equations

$$\alpha^2 + \beta^2 + \gamma^2 = -10 - 12i$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = 5 + 6i$$

- (a) Show that $\alpha + \beta + \gamma = 0$.

(2)

- (b) The numbers α , β and γ are also the roots of the equation

$$z^3 + pz^2 + qz + r = 0$$

Write down the value of p and the value of q .

(2)

- (c) It is also given that $\alpha = 3i$.

- (i) Find the value of r .

(3)

- (ii) Show that β and γ are the roots of the equation

$$z^2 + 3iz - 4 + 6i = 0$$

(2)

- (iii) Given that β is real, find the values of β and γ .

(3)

(Total 12 marks)

Q7.

It is given that $z = x + iy$, where x and y are real numbers.

- (a) Find, in terms of x and y , the real and imaginary parts of

$$i(z + 7) + 3(z^* - i)$$

(3)

- (b) Hence find the complex number z such that

$$i(z + 7) + 3(z^* - i) = 0$$

(3)

(Total 6 marks)

Q8.

- (a) It is given that $z_1 = \frac{1}{2} - i$.

- (i) Calculate the value of z_1^2 , giving your answer in the form $a + bi$.

(2)

- (ii) Hence verify that z_1 is a root of the equation

$$z^2 + z^* + \frac{1}{4} = 0$$

(2)

(b) Show that $z_2 = \frac{1}{2} + i$ also satisfies the equation in part (a)(ii). (2)

(c) Show that the equation in part (a)(ii) has two equal **real** roots. (2)

(Total 8 marks)

Q9.

(a) Show that $(1 + i)^3 = 2i - 2$. (2)

(b) The cubic equation

$$z^3 - (5 + i)z^2 + (9 + 4i)z + k(1 + i) = 0$$

where k is a real constant, has roots α , β and γ .

It is given that $\alpha = 1 + i$.

(i) Find the value of k . (3)

(ii) Show that $\beta + \gamma = 4$. (1)

(iii) Find the values of β and γ . (5)

(Total 11 marks)

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Q10.

It is given that $z = x + iy$, where x and y are real.

(a) Find, in terms of x and y , the real and imaginary parts of

$$(z - i)(z^* - i)$$

(3)

(b) Given that

$$(z - i)(z^* - i) = 24 - 8i$$

find the two possible values of z .

(4)
(Total 7 marks)

Q11.

The complex number z is defined by

$$z = 1 + i$$

- (a) Find the value of z^2 , giving your answer in its simplest form. (2)
- (b) Hence show that $z^8 = 16$. (2)
- (c) Show that $(z^*)^2 = -z^2$. (2)
- (Total 6 marks)**

Q12.

It is given that $z = x + iy$, where x and y are real numbers.

- (a) Find, in terms of x and y , the real and imaginary parts of $(1 - 2i)z - z^*$ (4)
- (b) Hence find the complex number z such that $(1 - 2i)z - z^* = 10(2 + i)$ (2)
- (Total 6 marks)**

Q13.

The complex number z is defined by

$$z = x + 2i$$

where x is real.

- (a) Find, in terms of x , the real and imaginary parts of:
- (i) z^2 ; (3)
- (ii) $z^2 + 2z^*$. (2)
- (b) Show that there is exactly one value of x for which $z^2 + 2z^*$ is real. (2)
- (Total 7 marks)**

Q14.

It is given that $z_1 = 2 + i$ and that z_1^* is the complex conjugate of z_1 .

Find the real numbers x and y such that

$$x + 3iy = z_1 + 4iz_1^*$$

(Total 4 marks)

Q15.

- (a) (i) It is given that α and β are the roots of the equation

$$x^2 - 2x + 4 = 0$$

Without solving this equation, show that α^3 and β^3 are the roots of the equation

$$x^2 + 16x + 64 = 0$$

(6)

- (ii) State, giving a reason, whether the roots of the equation

$$x^2 + 16x + 64 = 0$$

are real and equal, real and distinct, or non-real.

(2)

- (b) Solve the equation

$$x^2 - 2x + 4 = 0$$

(2)

- (c) Use your answers to parts (a) and (b) to show that

$$(1 + i\sqrt{3})^3 = (1 - i\sqrt{3})^3$$

(2)

(Total 12 marks)

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Q16.

It is given that $z = x + iy$, where x and y are real numbers.

- (a) Find, in terms of x and y , the real and imaginary parts of

$$3iz + 2z^*$$

where z^* is the complex conjugate of z .

(3)

- (b) Find the complex number z such that

$$3iz + 2z^* = 7 + 8i$$

(3)

(Total 6 marks)

Q17.

The cubic equation

$$z^3 + qz + (18 - 12i) = 0$$

where q is a complex number, has roots α , β and γ .

(a) Write down the value of:

(i) $\alpha\beta\gamma$; (1)

(ii) $\alpha + \beta + \gamma$. (1)

(b) Given that $\beta + \gamma = 2$, find the value of:

(i) α ; (1)

(ii) $\beta\gamma$; (2)

(iii) q . (3)

(c) Given that β is of the form ki , where k is real, find β and γ . (4)

(Total 12 marks)

Q18.

(a) Solve the following equations, giving each root in the form $a + bi$:

(i) $x^2 + 16 = 0$; (2)

(ii) $x^2 - 2x + 17 = 0$. (2)

(b) (i) Expand $(1 + x)^3$ (2)

(ii) Express $(1 + i)^3$ in the form $a + bi$. (2)

(iii) Hence, or otherwise, verify that $x = 1 + i$ satisfies the equation

$$x^3 + 2x - 4i = 0$$

(2)

(Total 10 marks)

Q19.

The cubic equation

$$z^3 + 2(1 - i)z^2 + 32(1 + i) = 0$$

has roots α , β and γ .

- (a) It is given that α is of the form ki , where k is real. By substituting $z = ki$ into the equation, show that $k = 4$.

(5)

- (b) Given that $\beta = -4$, find the value of γ .

(2)

(Total 7 marks)

Q20.

It is given that $z = x + iy$, where x and y are real numbers.

- (a) Find, in terms of x and y , the real and imaginary parts of

$$z - 3iz^*$$

where z^* is the complex conjugate of z .

(3)

- (b) Find the complex number z such that

$$z - 3iz^* = 16$$

(3)

(Total 6 marks)

Q21.

- (a) (i) Calculate $(2 + i\sqrt{5})(\sqrt{5} - i)$.

(3)

- (ii) Hence verify that $\sqrt{5} - i$ is a root of the equation

$$(2 + i\sqrt{5})z = 3z^*$$

where z^* is the conjugate of z .

(2)

- (b) The quadratic equation

$$x^2 + px + q = 0$$

in which the coefficients p and q are real, has a complex root $\sqrt{5} - i$.

- (i) Write down the other root of the equation.

(1)

- (ii) Find the sum and product of the two roots of the equation.

(3)

(iii) Hence state the values of p and q .

(2)

(Total 11 marks)

Q22.

The complex numbers z_1 and z_2 are given by

$$z_1 = \frac{1+i}{1-i} \quad \text{and} \quad z_2 = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

(a) Show that $z_1 = i$.

(2)

(b) Show that $|z_1| = |z_2|$.

(2)

(c) Express both z_1 and z_2 in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$.

(3)

(d) Draw an Argand diagram to show the points representing z_1 , z_2 and $z_1 + z_2$.

(2)

(e) Use your Argand diagram to show that

$$\tan \frac{5}{12}\pi = 2 + \sqrt{3}$$

(3)

(Total 12 marks)

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Q23.

It is given that $z = x + iy$, where x and y are real numbers.

(a) Write down, in terms of x and y , an expression for

$$(z + i)^*$$

where $(z + i)^*$ denotes the complex conjugate of $(z + i)$.

(2)

(b) Solve the equation

$$(z + i)^* = 2iz + 1$$

giving your answer in the form $a + bi$.

(5)

(Total 7 marks)

Q24.

The cubic equation

$$z^3 - 4iz^2 + qz - (4 - 2i) = 0$$

where q is a complex number, has roots α , β and γ .

(a) Write down the value of:

(i) $\alpha + \beta + \gamma$;

(1)

(ii) $\alpha\beta\gamma$.

(1)

(b) Given that $\alpha = \beta + \gamma$, show that:

(i) $\alpha = 2i$;

(1)

(ii) $\beta\gamma = -(1 + 2i)$;

(2)

(iii) $q = -(5 + 2i)$.

(3)

(c) Show that β and γ are the roots of the equation

$$z^2 - 2iz - (1 + 2i) = 0$$

(2)

(d) Given that β is real, find β and γ .

(3)

(Total 13 marks)

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Q25.

It is given that $z = x + iy$, where x and y are real numbers.

(a) Write down, in terms of x and y , an expression for z^* , the complex conjugate of z .

(1)

(b) Find, in terms of x and y , the real and imaginary parts of

$$2z - iz^*$$

(2)

(c) Find the complex number z such that

$$2z - iz^* = 3i$$

(3)

(Total 6 marks)

Q26.

The equation

$$x^2 - 4x + 13 = 0$$

has roots α and β .

- (a) (i) Write down the values of $\alpha + \beta$ and $\alpha\beta$. (2)
- (ii) Deduce that $\alpha^2 + \beta^2 = -10$. (2)
- (iii) Explain why the statement $\alpha^2 + \beta^2 = -10$ implies that α and β cannot both be real. (2)
- (b) Find in the form $p + iq$ the values of:
- (i) $(\alpha + i) + (\beta + i)$; (1)
- (ii) $(\alpha + i)(\beta + i)$. (2)
- (c) Hence find a quadratic equation with roots $(\alpha + i)$ and $(\beta + i)$. (2)

(Total 11 marks)